

New Tools & Reused Old Tools

New Tools for AI

- The GUI days are gone, the CUI days are back.
- We don't click buttons anymore; instead, we use CLI to interact with AI actively.
- With AI, it's like working with multiple software engineers at the same time, and we need a CLI (not a GUI) for this goal.

Old Tools are back

- tmux is widely used with Claude code
- vim (nvim) is widely used to give inputs quickly and effectively.
- Git worktree is used for parallel working with AI (isolation among branches).
- VS Code Multi-root Workspace is used for managing multiple workspaces (by AI).

New Approaches of using AI

Prompt: What should the AI do?

Context: What should the AI know?

Harness: How should the AI operate?

Prompt → instructions

Context → information

Harness → orchestration

Harness Engineering

- With the Anthropy source code leaks, we can guess why Anthropic LLM gives better answers.
- It can orchestrate additional tools to get the answers better and quicker.
- In other words, they know how to solve problems using AI effectively, and it is called "Harness Engineering".

The Future of AI Engineering

- It's likely to be using RAG for Harness Engineering.
- It's likely to use local LLM & RAG & Ontology for (relatively) simple and secure work.
- We will purchase LLM/AI computing power when we need higher-level design & architecture.

We need to understand how LLM works

- There are many (amazingly good) resources on the web.
- Anyone can learn how to build an LLM from scratch.
- Building and using LLM can be common sense very soon.
- Building and using RAG can be the most fundamental skills for problem solvers/system builders.

Building the 2nd Brain

- It's not about the tools, it's about how you use your brain to solve problems.
- It's a good idea to start with notes and pens to understand how you learn the best
- We started with the idea I & C (Invite & Control) when we use AI.
- Now, its' ACC (accelerate)

ACC

- Automate
 - Don't do things manually when you can do automate.
 - Ask LLM how to automate
 - Master script languages such as shell/python/ruby for automation

- Context & Connect
 - Connect ideas, concepts, patterns, or anything.
 - The connection should be context based

So, when we build our 2nd brain, we need to build to make ACC possible using the system.

The most important tools for 2nd Brain (ACC approach)

- Git/GitHub
 - You should be able to revert anything back.
- Editor
 - You should be able to add/delete/manipulate the content as quickly, effectively, and easily as possible.
 - nvim is gaining more and more popularity.
 - any tool is OK as long as (1) it's light and fast and (2) powerful enough to do complicated things.

Then try these

- Obsidian
 - Text based (markdown and YAML)
 - The markdown/YAML file is database content
 - Powerful 3rd party tools (plugins)
 - JavaScript (Electron) based, so you can make your own plugins easily
 - Cross-platform
 - Customizable for any purpose you need
 - ...

- NVIM (neo vim)
 - Light & Fast
 - Based on the good old vim
 - Powerful 3rd party tools (plugins)
 - Lua based, so you can make your own plugins easily
 - Cross-platform
 - Customizable for any purpose you need
 - ...

- UNIX Shell script
 - For Linux users, it's built-in
 - For Mac users, it's built-in
 - For windows users, use WLS2

Anything can (should be) be automized using anytools
(Git/Obsidian/NVIM/Shell ... with AI)

New AI Tools

- LLM Wiki (LLM makes Wiki for you)
 - AI legendary Karpathy
 - <https://gist.github.com/karpathy/442a6bf555914893e9891c11519de94f>
- Mem Palace (AI memory system)
 - Actress Yoyovich (Resident evil actor) created this
 - <https://github.com/MemPalace/mempalace>

Old Ideas but Important

- MCP
- RAG
- Ontology
- ...

You can learn anything with AI as long as you don't know about them.